Preliminary Sample-Size Analysis for CLAPS

Pablo Bernabeu and CLAPS collaborators

3 September 2026

Abstract

CLAPS asks whether the acceptability of a passive sentence depends on how strongly the event affects the person or thing it happens to, and whether that pattern holds across languages. This report asks how large a sample is needed to answer that convincingly. The answer turns on an unusual feature of the design. Affectedness is a property of the verb, not of the participant, so precision is governed by how many verbs are tested. Past a point, recruiting more participants stops helping.

At the 72 verbs used here, that limit binds hardest on the design as first specified, which asked for both focal predictions in the same study. English's ceiling under that rule is 81% however many people are recruited, held down by the affectedness slope for passives, which tops out at 83%. Judged on the active-by-affectedness interaction alone, the prediction the theory turns on, the ceilings are 97.1% for English and 99.7% for Norwegian. Turkish is the exception under every rule, its interaction ceiling being 65%. Earlier, more optimistic simulations had assumed less variation between verbs than the pilot data show.

The recommendation is a confirmatory test in each language, judged on the interaction alone at a Bayes factor of 10, with 80 participants per language and the current verb pool. At that sample, English detects the interaction in 90% of simulated studies [0.83, 0.95] and Norwegian in 97% [0.92, 0.99]. The English figure meets a 90% target as a point estimate, and its plateau above 80 participants sits just below it, at 89%. Turkish reaches 57% and is reported as estimation. The three languages analysed together remain in the preregistration as the cross-linguistic synthesis, while the confirmatory claim rests on the per-language tests. With three languages, that model succeeds in 57% [0.47, 0.67] of simulated studies at a Bayes factor of 10, and no sample size lifts it. The threshold matters because this Bayes factor is directional: 3 corresponds to a one-sided tail probability of 0.25 and 6 to 0.14, both too lenient for a confirmatory claim under review, while 10 is 0.091 on that scale and stricter once the prior's shrinkage is allowed for. The verb pool cannot grow and the sample is fixed in advance.

This note reports preliminary sample-size guidance for CLAPS (A Crosslinguistic Approach to Passive Semantics). The guidance comes from a Bayes-factor design analysis. A design analysis of this kind simulates the planned model many times under plausible true effects, then asks how often a given sample size would yield compelling evidence for the focal predictions.

Two versions of the analysis are run. The primary one is data-grounded. It fits the planned model to each language's real pilot data and simulates new studies from that fit, so its results reflect the effects observed in the CLAPS pilot. The collaborators asked the design analysis to use that pilot as its basis. A single pilot's point estimate is noisy and tends to overstate power, the concern that Albers and Lakens (2018) raise for pilot-based power analyses. The data-grounded analysis avoids that point estimate in two ways. First, on each simulated study it draws the focal effects from the pilot posterior, the range of effect sizes the pilot data still leave plausible, so that the reported probability of a decisive Bayes factor reflects the pilot's uncertainty about the size of the effect. This yields an assurance, a power averaged over the plausible sizes of the effect. Second, it also evaluates a deliberately conservative lower bound on the effect, as a safeguard against an optimistic pilot. A second version, for comparison, is based on the pooled effect sizes from the five previously published studies. It provides an independent cross-check against the broader evidence base.

The two versions agree on the central structural point. The power to detect the affectedness effect is governed by the number of verbs in the materials, because affectedness is a property of the verb. They diverge on whether a modest sample is enough. On the published-literature effects, a per-language sample of at most 50 participants would suffice at the full verb set. On the pilot effects, and with the verb-level variability measured from the pilot instead of assumed, the picture is less favourable. On both focal predictions together, the rule the design first specified, Norwegian reaches the 80% target at about 100 participants, but English and Turkish do not reach it at any simulated sample size. For those two languages, the binding constraint is the number of verbs, and in Turkish also the strength of the effect itself, so adding participants alone cannot bring the design to the usual power target. Because the data-grounded analysis is the agreed primary basis, this is the finding that should guide the design. The report quantifies these limits analytically, explains why the two analyses disagree, and sets out design options that remain feasible for a volunteer consortium. The recommendation is a confirmatory test in each language on the interaction alone, with the pooled cross-linguistic model preregistered as the synthesis, and the sections below give the figures behind both.

The data-grounded analysis is substantially complete. Results are in for all three languages up to 130 participants, for English up to 400 and for Turkish up to 400. Nothing further is being simulated: the project has capped recruitment at 100 participants, so the cells above that are of record interest only. Each simulated condition rests on 22 to 144 studies, so the individual figures carry a Monte-Carlo error of several percentage points, that is, the uncertainty that comes from basing each figure on a limited number of simulated studies. The 400-participant simulations now confirm the English plateau, the levelling-off of power short of the target, so it no longer rests on extrapolation. The recommendation below is final on this evidence, and no result is outstanding. The refit described under the pooled model returned on 27 August, and it bears on the pooled figures alone.

Background and Motivation

The question CLAPS addresses is whether the acceptability of passive and related sentence types tracks the semantic affectedness of the patient, and whether that relationship differs by sentence type. Affectedness, the degree to which a patient or theme undergoes a change of state or is causally impinged upon by the action, is a core dimension of argument structure (Beavers, 2011; Dowty, 1991). It has long been argued to be the semantic heart of the passive, which carries the meaning that the surface subject is in a state or circumstance characterised by the agent having acted upon it (Pinker et al., 1987). Across languages, adult acceptability-judgement and comprehension studies support the view that passive acceptability is semantically structured rather than purely syntactic (Ambridge et al., 2016; Ambridge et al., 2023; Aryawibawa & Ambridge, 2018; Bidgood et al., 2020; Darmasetiyawan & Ambridge, 2022; Liu & Ambridge, 2021).

The cross-linguistic evidence base to date spans English, Indonesian, Balinese, Hebrew and Mandarin. A Bayesian meta-analytic synthesis finds a consistent pattern across them (Ambridge et al., 2023). Affectedness raises acceptability for both passives and actives, more strongly for passives, and does not raise it for pseudo-passives. The affectedness constraint has not yet been tested in Turkish or Norwegian. CLAPS extends the paradigm to these two typologically and morphologically distinct languages. No prior language-specific effect-size estimates exist for them, so the CLAPS pilot supplies the first such estimates. The data-grounded analysis uses these pilot estimates as its primary basis, and the cross-linguistic literature provides an independent comparison, which turns out to be the more optimistic of the two.

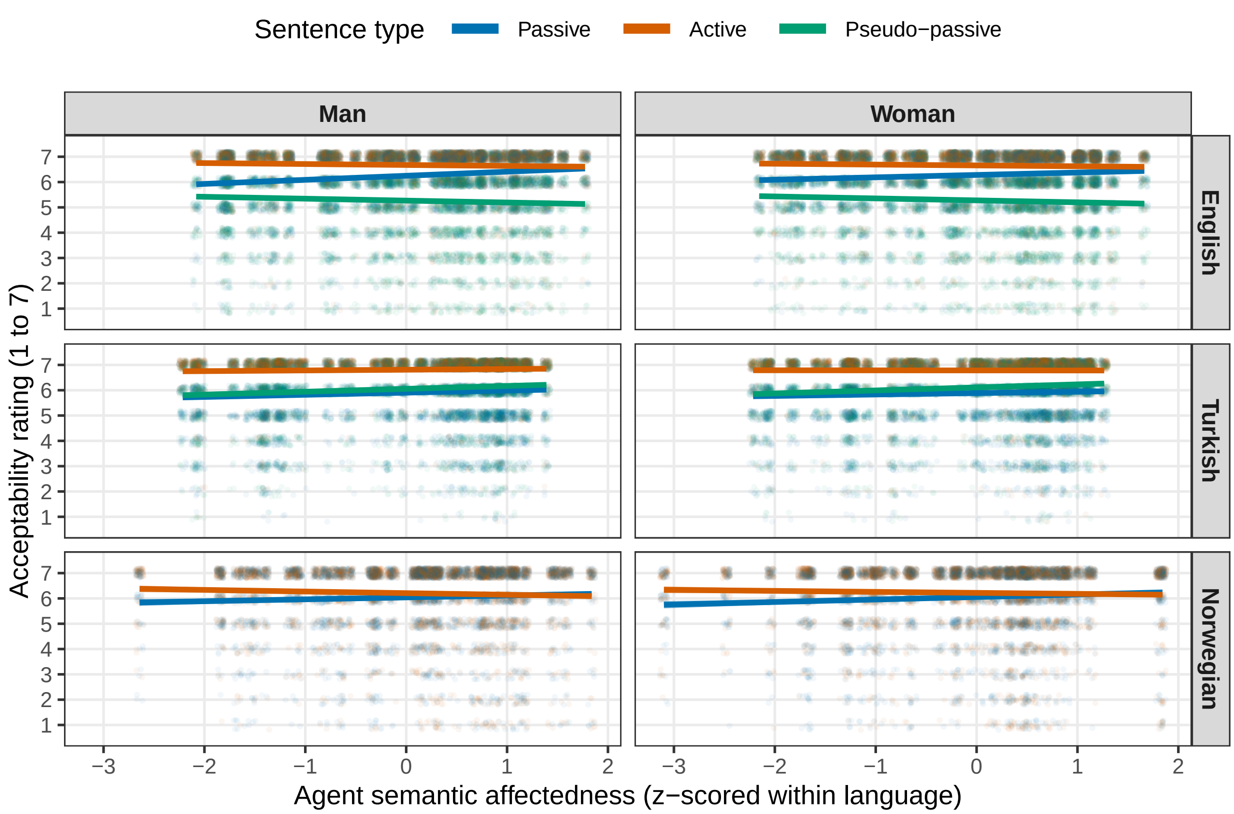
The Acceptability Model and the Focal Predictions

Each acceptability judgement is a discrete rating from 1 to 7. These ratings are modelled with a Bayesian cumulative-logit mixed-effects model, an ordinal model that respects the discreteness, ordering and bounded range of rating data while leaving the underlying scale free to vary across conditions (Bürkner & Vuorre, 2019). Treating the same ratings as continuous, under a Gaussian likelihood, biases the regression coefficients and inflates error rates where floor or ceiling effects are present (Liddell & Kruschke, 2018). Recent empirical work shows that modelling them cumulatively recovers psycholinguistic effects more accurately (Taylor et al., 2023; Veríssimo, 2021).

The expected pattern is already visible in the pilot data, drawn from the sample whose composition by language is given in Table 4 in Appendix A. Figure 1 shows the raw acceptability ratings against agent semantic affectedness, separately by sentence type, referent gender and language. The affectedness slope is positive and steepest for the passive in English and Norwegian, and shallower for the active and the pseudo-passive. The trends are very similar for male and female referents within each language. This is reassuring, because the focal effects should not be driven by a referent-gender imbalance.

Figure 1

Raw Pilot Acceptability by Agent Affectedness, Sentence Type and Referent Gender



Note. Panels are by language across rows and by referent gender across columns. Points are individual pilot ratings, jittered, and lines are per-sentence-type linear trends with 95% confidence bands. Affectedness is z-scored within language. The pilot data are used for calibration only. Norwegian has no pseudo-passive.

The model predicts acceptability from sentence type, from verb affectedness, and from their interaction. The passive is the reference level, and the active and the pseudo-passive are the comparisons. The focal predictions concern the affectedness slope in the passive and how it changes across sentence types. The primary prediction (H1b) is one-tailed and holds that affectedness raises passive acceptability more than it raises active acceptability, so the active-by-affectedness interaction is negative. A supporting prediction (H1a) holds that affectedness raises acceptability in the passive itself, so the passive affectedness slope is positive. The confirmatory decision rests on H1b alone. H1a is a prespecified positive control: it checks that the interaction means what it is meant to mean, and is reported as an estimate. An earlier version of the design required both predictions in the same study, and the results below are what led to dropping that requirement. A secondary, two-tailed prediction (H2) concerns the pseudo-passive-by-affectedness interaction. It does not gate the recommendation and does not apply to Norwegian, which has no pseudo-passive.

Design-Analysis Method

The design analysis estimates statistical power by simulation. Each simulated dataset is drawn from the same cumulative-logit model that will be used for analysis, under assumed values for the effects of interest. Each dataset is then analysed in the same way. The proportion of simulated studies that yield decisive evidence estimates the power at that sample size. The whole procedure stays within the Bayesian framework. This matters because the confirmatory test is a Bayes factor: the power to return a decisive Bayes factor is not the same quantity as the power to return a significant p-value.

Each verb carries a single affectedness value, held constant across participants and sentence types, exactly as in the real materials. The number of verbs therefore determines the precision with which the affectedness effect can be estimated. The verb count is treated as a design dimension alongside the participant count.

The two analyses differ in where they take the assumed true effects and the affectedness values. The data-grounded analysis, which is primary, fits the planned model to each language's pilot data. It then takes the population effects, the random-effect covariance and the ordinal thresholds from that fit, and it reads each verb's actual affectedness score from the pilot data. The literature-anchored cross-check instead sets the population effects to values from the cross-linguistic meta-analysis. On the standardised affectedness predictor, the passive affectedness slope is set near the pooled value of about 0.23 logits per standard deviation (a logit is a unit of the model's underlying scale, not a raw point on the 1 to 7 rating), with a negative active-by-affectedness interaction of about -0.15 and a small positive pseudo-passive-by-affectedness interaction. In the cross-check, affectedness is spread across a plausible range, and the by-participant and by-verb random-effect standard deviations are set to realistic magnitudes. The referent-gender covariate is held out of both preliminary runs and is examined separately.

The analysis model uses the maximal random-effects structure consistent with the design, meaning the most complex structure the design supports, so any effect that can vary across participants or verbs is allowed to do so. It has correlated by-participant slopes for sentence type, affectedness and their interaction, together with by-verb intercepts and sentence-type slopes (Barr et al., 2013). Because affectedness does not vary within a verb, a by-verb affectedness slope cannot be estimated and is omitted, so affectedness enters as a population-level effect, one estimated for the sample as a whole. The priors are weakly to moderately informative and centred at zero on the focal slopes, so that the directional tests are not pre-loaded. A normal prior of standard deviation 0.5 is placed on the affectedness slope and on the active interaction, and a slightly wider normal prior of 0.6 on the pseudo-passive interaction. The sentence-type main effects take a wider normal prior. Student-t priors are placed on the ordinal thresholds and on the by-participant and by-verb standard deviations, and an LKJ prior, the standard prior for a correlation matrix, on the correlations among those random effects (Gelman, 2006; Lewandowski et al., 2009).

Evidence for each prediction is summarised by a directional Savage-Dickey Bayes factor, a number that weighs how strongly the data favour the predicted direction of the effect over the opposite direction, obtained by sampling the prior alongside the posterior (Wagenmakers et al., 2010). A simulated study counts as a detection when its Bayes factor reaches 10, the threshold the project set for strong evidence. The Bayes factor used here is directional, meaning the posterior odds that the coefficient carries the predicted sign divided by the same odds under the prior, so under a symmetric prior a value of 10 is odds of 10 to 1 on the direction. That is a posterior probability of 0.91, or about 1.34 posterior standard deviations from zero. It is not the point-null Bayes factor of a default two-sided test, which is a different quantity on a different scale, and under the same $N(0, 0.5)$ prior a directional 10 is worth a point-null 1 at each language's precision. The frequentist equivalent is larger than 1.34, because the zero-centred prior shrinks the estimate towards zero, so the sampling estimate has to sit further from zero than the posterior does, by a factor of $\sqrt{1 + \mathrm{SE}^2/0.5^2}$. For English at 80 participants, where the simulated detection rate implies under a normal approximation a standard error of about 0.34 on the model scale, a Bayes factor of 10 needs an estimate about 1.61 standard errors from zero, close to the 1.64 of a one-sided $p < .05$. For Norwegian and Turkish, whose standard errors are smaller, the equivalent is nearer 1.41, a one-sided tail of about .08. Power is the proportion of simulated studies that reach it. Results are reported at Bayes factors of 10, 6 and 3 throughout, since the stored per-cell Bayes factors allow any threshold to be scored without re-running anything. The recommendation, set out below, is judged on the interaction alone at a Bayes factor of 10, and fixes a single sample size in advance instead of taking the smallest one that meets a power target.

Each replicate is fitted with four Markov chains of 3,000 iterations, 1,000 of them warm-up, at the same target acceptance rate of 0.99 and maximum tree depth of 12 that the confirmatory analysis will use, which leaves 8,000 post-warm-up draws. The proposal specifies a heavier run for that single confirmatory fit, sixteen chains of 5,000 iterations with 2,000 of warm-up, or 48,000 draws. The shorter chains add Monte-Carlo noise to each simulated Bayes factor without shifting it systematically, so the power figures below are, if anything, mildly conservative.

The literature-anchored grid crosses the three languages with participant counts from 50 to 120. The output read here covers English, Norwegian and Turkish from 50 to 120 participants at verb counts of 40 and 72. Each completed cell holds 50 replicates, giving a Monte-Carlo standard error of about five to six percentage points near a power of 80%, so estimates close to the threshold are not perfectly ordered across adjacent sample sizes. The central code for the data-generating process, the model and the Bayes factor is reproduced in Appendix B. The full operating characteristics of this run, cell by cell, are reported in Table 5 in Appendix C, and Appendix D maps the whole repository, including which file implements each of the two simulation engines described above and how the pooled analysis reuses one of them.

Results

Data-grounded results

The data-grounded analysis is the primary basis for the recommendation, and it is now substantially complete. In its assurance form, it draws the focal effects afresh from each language's pilot posterior on every simulated study, so the reported power carries the pilot's own uncertainty about the size of those effects. Table 1 reports the joint power, the proportion of simulated studies in which both focal predictions clear the Bayes-factor threshold together, at the pilot's real verb count.

Table 1

Preliminary Data-Grounded Joint Power by Participant Count and Language, Assurance Variant

Participants	English	Turkish	Norwegian
30	25.0%	20.8%	41.7%
50	41.7%	37.5%	66.7%
70	47.9%	36.8%	68.8%
80	60.0%	41.7%	68.3%
100	55.6%	38.2%	81.9%
130	45.8%	29.2%	87.5%
150	62.5%	35.0%	-
200	65.0%	42.5%	-
250	70.0%	42.5%	-
300	72.5%	35.9%	-
400	62.5%	40.9%	-

Note. Joint power is the proportion of studies in which both focal predictions clear a Bayes factor of 10 together, at 72 verbs and 66 for Norwegian. Cells holding fewer than 10 replicates are excluded, since they cannot support the precision reported here. The rest hold 22 to 144 studies each. Percentages are shown to one decimal place to preserve the exact Monte-Carlo summaries. Norwegian at 130 participants sits a little above the approximate infinite-participant ceiling in Table 2, at $21/24 = 87.5\%$ against a Monte-Carlo standard error of 6.8%, so the difference is sampling variation and the two agree.

The pattern differs sharply by language. Norwegian reaches the 80% target at about 100 participants and is comfortably powered beyond it. English and Turkish do not reach it at any simulated sample size. For English, the analytic ceiling is 81%, but that is the limiting value with unlimited participants, and its simulated joint power remains below 80%. It is 72.5% by 300 participants and 62.5% at 400. The reason is structural. The joint requirement is held back by the affectedness main effect (H1a), whose detection rate levels off at about three in four, while the interaction (H1b) climbs comfortably past it. Because affectedness is a property of the verb, its main effect is estimated from the 72 verbs, so adding participants does not sharpen it. The precision of that effect, and with it the power to detect it, is set by the number of verbs. For Turkish, both focal effects are weak in the pilot. The joint power never exceeds 42.5% at any simulated sample size, and the present design does not bring it to target.

A conservative safeguard variant, which fixes each focal effect at a cautious lower bound of its pilot posterior, is lower again. Its highest joint power over the samples run is 58% for Norwegian, 25% for English and 12% for Turkish. This is the design's power under a deliberately pessimistic reading of the pilot.

The pilot posterior itself puts the probability that both focal effects have their predicted sign at 99.6% for English, 99.5% for Norwegian and 94.0% for Turkish, so the pilot points the right way in all three languages. The shortfall for English and Turkish is therefore a matter of precision. Because affectedness varies only between verbs, the standard error of its effect cannot fall below the by-verb variability divided by the spread of the affectedness scores, however many participants respond. This is the stimulus-sampling limit long recognised in psycholinguistics (Clark, 1973) and quantified for designs with crossed random effects by Westfall et al. (2014; see also Judd et al., 2017; Brysbaert & Stevens, 2018). The limit can be computed directly from the fitted pilot models, with no simulation involved. Table 2 reports the resulting upper bounds, the detection rates that would remain even with unlimited participants at the current verb counts.

Table 2

Analytic Upper Bounds on Detection with Unlimited Participants, at the Current Verb Counts

Language	H1a ceiling	H1b ceiling	Joint ceiling	Verbs for 80%	Verbs for 90%
English	83%	97%	81%	72	124
Turkish	77%	65%	49%	348	> 1,200
Norwegian	85%	100%	85%	66	98

Note. Ceilings are the probability of a Bayes factor of at least 10 with unlimited participants, from the 8,000 pilot posterior draws and the fitted by-verb covariance at 72 verbs and 66 for Norwegian. The H1b column is the active-by-affectedness interaction, the confirmatory prediction, and the H1a column the passive affectedness slope. The last columns give the verbs needed for an 80% and 90% joint ceiling. A ceiling just above target, as for English, still leaves it out of reach, since any finite sample sits below its ceiling. The English interaction ceiling of 97.1% is likewise the limit of an approximate formula: over the 480 replicates at 80 participants and above, the interaction is detected in 89% [0.86, 0.91], and the gap is the formula's, which ignores some of the covariance among the by-verb effects.

These bounds agree closely with the simulations. English's simulated joint power climbs to 72.5% by 300 participants against a ceiling of 81%. Turkish never exceeds 42.5% against a ceiling of 49%. Norwegian's simulated 81.9% at 100 participants matches its ceiling of 85% within Monte-Carlo error. The agreement between two entirely different routes to the same quantities, stochastic simulation and direct calculation, is a strong check that the pipeline is sound. A further check makes the point starkly for English. The pilot's own posterior standard deviation for the affectedness slope, a measure of how precisely the pilot pins that slope down, is 0.19, already close to the verb-count floor of 0.19. A replication with the same 72 verbs would therefore have little room to estimate that slope more precisely than the pilot already has, whatever its sample size. The residual gap reflects the floor formula's own approximation, since it ignores some of the covariance in the by-verb random effects.

The ceilings also show what would and would not help. Relaxing the evidence criterion from a Bayes factor of 10 to 6 or 3 lifts the English joint ceiling to 86% and 92%, but lifts Turkish only to 58% and 70%, so a weaker threshold cannot rescue the Turkish test. Adding verbs raises every ceiling, because the precision floor shrinks with the square root of the verb count. English would reach a joint ceiling of 90% at about 124 verbs, and Norwegian at about 98. Turkish would need about 348 verbs merely to reach a ceiling of 80%, beyond what a single language can supply, so for Turkish the remedy lies elsewhere, as taken up under Paths Forward below.

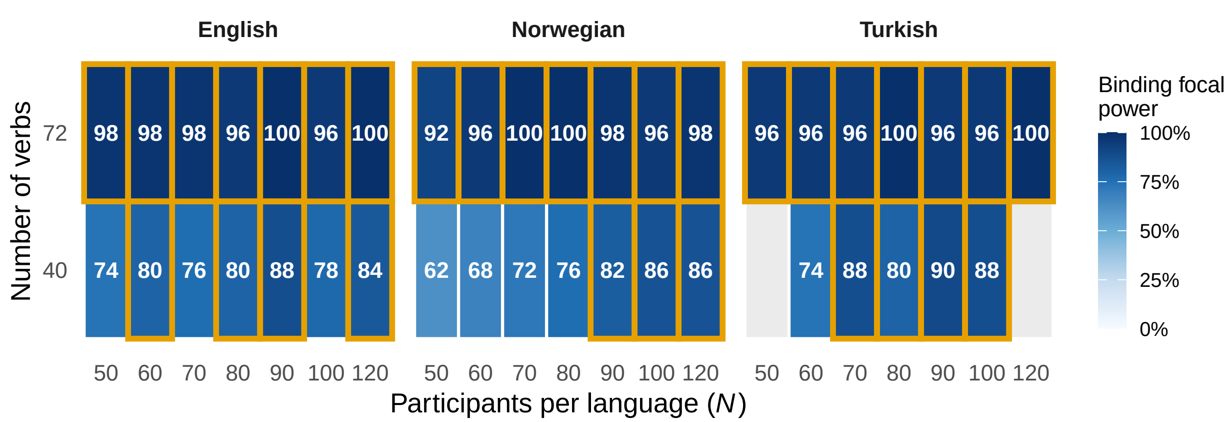
These results are preliminary in one remaining respect, since each condition rests on 22 to 144 simulated studies and so carries a Monte-Carlo error of a few percentage points at the better-populated sample sizes and more at the sparser ones. The English plateau itself, though, is now confirmed by the cells that have run. At 400 participants, with 32 replicates in, the joint power sits at 62.5% (20/32), compared with 72.5% (29/40) at 300 participants. These sparse estimates are not literally flat, but their decline is compatible with Monte-Carlo variation. Overturning the plateau would have required a sustained rise at 400 and 500, and the 400-participant data already rule that out. No further cells are being simulated, and by that test the sample sizes already run settle the plateau.

Literature-anchored cross-check

The grid is read as a surface over participant count and verb count, summarised by the binding focal power, the smaller of the two focal detection probabilities, since the design as first specified required both to clear 80%. The recommendation now gates on the interaction alone, so this surface records the earlier criterion. This cross-check is the more optimistic of the two analyses, because it fixes the effects at the pooled literature values and carries no uncertainty about them. Figure 2 shows this surface for all three languages. Cells outlined in gold meet the target, and the blank cells are Turkish at 40 verbs, never run at 120 participants and holding too few replicates at 50 to report. The pattern is clear. Power increases steeply with the number of verbs and only weakly with the number of participants.

Figure 2

Binding Focal Power by Participant Count and Verb Count, for Each Language

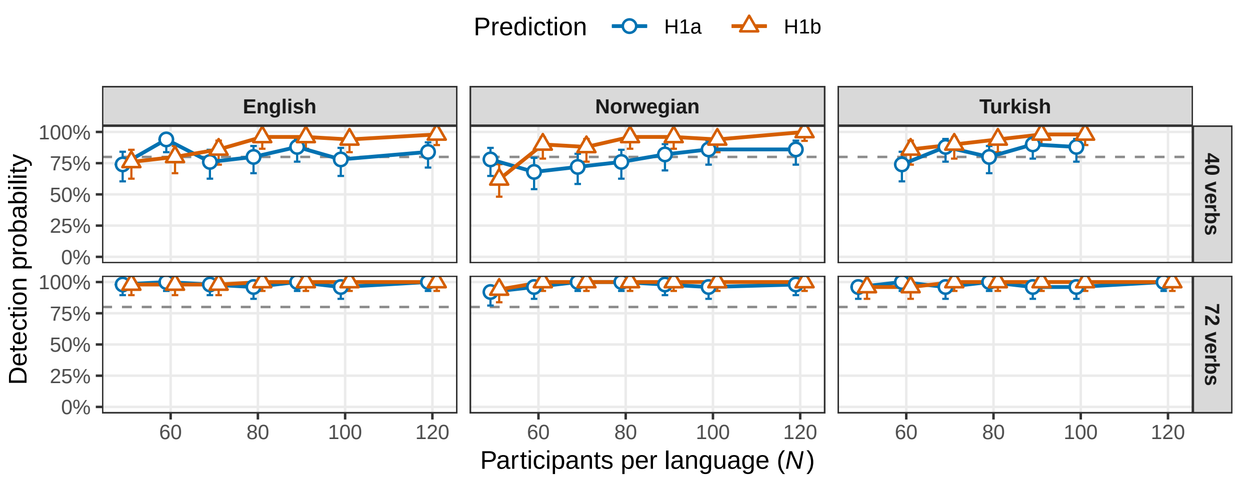


Note. Each cell gives the binding focal power as a percentage, with darker shading for higher power. Cells outlined in gold meet the earlier 80% target. Blank cells were either not run or hold too few replicates to report.

The two axes are not equally important. At 40 verbs in English, the binding power is 74% at 50 participants and reaches the 80% target only between about 60 and 90 participants, within Monte-Carlo noise. At 72 verbs, it is already 92% to 98% at 50 participants. Increasing the number of verbs from 40 to 72 raises power more than increasing the number of participants does, as follows from affectedness being a verb-level property. Figure 3 separates the two focal predictions and shows how each approaches the target as the sample grows, at each verb count.

Figure 3

Detection Probability for the Focal Predictions by Verb Count and Language



Note. Points are detection probabilities over the simulated studies in each complete cell, with Wilson 95% intervals reflecting Monte-Carlo uncertainty. The dashed line marks the earlier 80% target. Points are dodged so the two predictions do not overlap. H1b is the active-by-affectedness interaction, which is the confirmatory prediction, and H1a the passive affectedness slope, which is reported as a positive control.

In the literature-anchored output, which covers English, Norwegian and Turkish, the binding focal power at the full 72-verb set is already between 92% and 98% at 50 participants, the smallest sample simulated, in every language included in this cross-check. On the literature effects, the adequate per-language sample is therefore at most 50 participants at the full verb set. This is the optimistic reading. The data-grounded results above, which condition on the weaker pilot effects, do not reach the target for English or Turkish at any sample size. Because 50 is the smallest sample in the main grid, the analysis places the adequate sample at or below 50 without determining it more precisely. Pinning it down further, below 50 participants, is not pursued here, since the recommendation rests on the data-grounded analysis.

Why the Two Analyses Disagree

The gap between the two analyses calls for an explanation, because the literature-anchored grid reports 92% to 98% at 50 participants on the same 72 verbs at which the data-grounded joint power never reaches 80%. Expressed per standard deviation of affectedness, the pilot's passive slope is 0.26 logits for English, 0.23 for Norwegian and 0.18 for Turkish, against a pooled literature value of about 0.23. The Turkish active interaction, at -0.15, matches the literature's -0.15 almost exactly, and the English interaction, at -0.44, is almost three times the literature value, so the CLAPS pilot effects are not unusually small.

The disagreement comes instead from the noise side of the simulation, in two specific assumptions. First, the literature-anchored simulator assumed a by-verb intercept standard deviation of 0.5, a measure of how much verbs differ in their baseline acceptability, where the fitted pilot models put it at 0.83 for English, 0.64 for Turkish and 0.66 for Norwegian. The precision floor on the affectedness slope scales directly with this quantity, so assuming 0.5 where the fitted value is 0.83 overstates the attainable evidence considerably. Second, the literature-anchored simulator included no by-verb sentence-type slopes at all, allowing no verb-to-verb variation in how sentence type shifts the ratings, where the fitted pilots put their standard deviation at 0.88 for English, 0.64 for Turkish and 0.34 for Norwegian. That omission left the interaction test with no verb-level noise floor in the simulation, which is why the earlier grid found the interaction so easy to detect. Both assumptions were reasonable defaults before the pilot models existed. The pilot has now measured the quantities they stood in for, and the data-grounded analysis uses the measured values. The earlier optimism, in short, came less from the assumed effect sizes than from understating how much verbs vary (Clark, 1973; Westfall et al., 2014).

Effect-Size Qualification

One assumption deserves particular care, because it bears directly on the absolute sample size. The numbers shown here condition on assumed effects set to the cross-linguistic pooled posterior means from a Bayesian meta-analytic synthesis of the passive-affectedness literature, which is a central estimate of the published effects (Ambridge et al., 2023). On the simulator's internal predictor scale, these appear as larger nominal values, but expressed per standard deviation of affectedness, they are about 0.23 and -0.15 logits, matching the pooled literature values. The larger nominal numbers therefore reflect the scaling alone. Two cautions nonetheless apply. The first is the one the previous section sets out. The fitted pilot models put the verb-level variability well above the values this grid assumed, and that difference, more than any difference in effect size, made the grid optimistic. The pilot effects themselves are broadly in line with the pooled literature values once both are expressed per standard deviation of affectedness, though the Turkish terms sit at the weak end, with the interaction at about -0.15 in those units. The second is that the literature is itself subject to publication bias and to the winner's curse, the tendency of published estimates, especially from small and low-powered studies, to overstate true magnitudes (Button et al., 2013; Gelman & Carlin, 2014; Ioannidis, 2008). Design analyses anchored to such estimates can therefore be optimistic about power (Albers & Lakens, 2018; Vasishth et al., 2018).

These cautions are the reason the data-grounded analysis is the primary basis for the recommendation. It draws the focal effects from each language's own pilot posterior and also evaluates a conservative lower bound. Conditioning in this way on the effects observed in CLAPS carries their uncertainty through to the reported power, and those results are reported above. They agree with this cross-check for Norwegian, powered at about 100 participants, and they diverge from it for English and Turkish, where the weaker pilot effects hold the joint power below target at every sample size run.

A separate point concerns the scale on which the prior sits, not the size of the assumed effect. The literature-anchored grid behind the figures expresses affectedness on the simulator's internal predictor, whose standard deviation is about 0.29. The real confirmatory analysis and the data-grounded analysis both place it on the analysis pipeline's Gelman scaling, which divides a predictor by twice its standard deviation and so leaves it with a standard deviation of 0.5, and the same weakly informative prior is applied throughout. A Bayes factor reflects the width of the prior relative to the coefficient it constrains. Holding the prior fixed while the predictor scale differs therefore leaves the literature-anchored grid mildly optimistic about the focal-slope Bayes factor, by a factor of about 1.7, relative to the analysis that will be run. The qualitative conclusion is unaffected, because the verb count dominates the result, the focal Bayes factors at the full verb set lie far above the threshold, and the data-grounded analysis already uses the correct scale. The cross-check grid was not re-run on the corrected scale and will not be, since the recommendation rests on the data-grounded analysis alone. The figures here should therefore be read as a mild upper bound on the Bayes-factor evidence.

For an effect that lives at the level of the participant, a weaker true effect simply needs a larger sample, and the rough guide is that a true effect around two-thirds of the assumed magnitude doubles the sample needed. The affectedness main effect lives at the level of the verb, however, where its precision is set by the number of verbs, so a weaker true effect cannot be recovered by recruiting more participants. This is exactly what the data-grounded results show. At the full verb count and the pilot's own effects, the affectedness power for English settles below the target and does not climb with sample size, and for Turkish neither focal effect is strong enough to bring the joint power to target. The effective levers are therefore the number of verbs, and for Turkish the strength of the interaction. Where feasible, the design should still be planned against a deliberately conservative, or safeguard, effect size (Perugini et al., 2014), and the data-grounded analysis builds this in directly through its assurance and safeguard variants. Discounting the literature-anchored grid towards weaker effects makes the same point: re-evaluated at 75% and 60% of the assumed effect magnitude, with the 60% level approximating the weaker effects seen in the pilot, it needs larger samples. The data-grounded analysis conditions on those pilot effects directly, so the primary results above already reflect that discounting.

Convergence and Computation

Convergence is sound across the grid, with no divergent transitions anywhere, which are the warning sign of an unreliable fit. At the smaller verb count, essentially every fit meets the strict criteria: an R-hat below 1.01, the check that the separate chains have settled on the same answer, together with bulk and tail effective sample sizes of at least 400, which measure how much independent information the draws carry. At 72 verbs, the maximal model occasionally falls just short of these criteria, in about one fit in ten at the smaller samples and up to one in five at the larger ones. The failures show up on the R-hat and effective-sample-size thresholds while divergences stay at zero, so they are a matter of how efficiently the sampler explores the posterior, and the specification itself stands. Restricting the power calculation to the fully converged fits leaves the estimates essentially unchanged. For English at 72 verbs and 50 participants, the binding focal power is 98% over all 50 fits and 98% over the 46 converged fits, so the recommendation does not depend on the difference. That describes the literature-anchored grid. The pilot-grounded replicates, which carry the fitted by-verb slopes and the larger variance components the pilot models estimate, pass the same screen far less often, 27 of 120 English fits at 80 participants, and the rates over the passing fits are reported beside the decision table below. There too the failures are on effective sample size and R-hat, with divergences and treedepth saturations at zero.

The simulations ran on the University of Oxford's ARC service, across its two clusters, under R 4.4.2 from the module `R/4.4.2-gfbf-2024a`. Each replicate is fitted with brms 2.21.0 through the cmdstanr backend, so the sampler is CmdStan 2.39.0. The reported fits were compiled by the gcc 13.3.0 that arrives with the R module, though the CmdStan installation now pins gcc 12.3.0 in its `make/local`, so a rerun would compile with that instead. The compiler is worth naming, because a Stan model is compiled C++ and the toolchain is part of what decides whether a fit reproduces exactly. The distribution recorded in `config/arc_modules.yaml` is the login node's, where the record was captured, while the compute nodes run an ARC-built kernel of the same family. Containers are not available on this cluster, which was checked in July 2026, so reproducibility rests on the module pin and the lockfile.

`renv.lock` in the repository records that environment. It fixes 176 packages at exact versions against R 4.4.2, each entry carrying the hash renv computes from the package's DESCRIPTION. All but one are pinned to CRAN, among them brms 2.21.0, rstan 2.32.7 and StanHeaders 2.32.10. The exception is cmdstanr 0.8.0, which CRAN does not distribute and which is locked from an r-universe mirror. The lockfile was checked against the cluster on 2 September 2026, when every version in it matched what was installed there. A companion record, `config/arc_environment_recorded.csv`, holds the wider set of 190 packages actually present, the same 176 with the fourteen base packages that come with R itself.

The report renders away from the cluster. Every power figure, Bayes factor, ceiling and convergence rate in it is read from the committed summary tables, with no replicate refitted at build time. The R that builds the document computes the descriptive pilot figure and its trend lines, the sample counts in Appendix A and the binomial intervals quoted in the text. Its version therefore bears on how the simulation results are presented, while the results themselves come from the cluster.

Secondary Prediction

The pseudo-passive-by-affectedness interaction reaches a detection probability of about 58% to 88% at 72 verbs. This remains below the focal predictions and reflects its smaller assumed magnitude, and the opposite-direction test is near zero throughout, as expected. This prediction is secondary and two-tailed, and it does not constrain the sample-size recommendation. Full operating characteristics for all four predictions, including the convergence rates, are reported in Appendix C.

Paths Forward

The analytic bounds turn the shortfall into a set of concrete design choices. Four options follow, in approximate order of increasing departure from the current plan. Two of them have since been closed by the project, and are kept here because the reasoning that ruled them out is part of the record.

The design is now capped at 80 participants per language, with 100 the absolute maximum any team would be asked for. The verb pool cannot grow beyond 72, because suitable verbs are scarce in every language and session length was never the constraint. The third option below is therefore unavailable. Collection must be at a single sample size fixed and announced in advance, because the teams have to take a stated number to their ethics committees, which rules out the fourth option. Many of the teams have not run a study of this type before and could not defend a more elaborate protocol to their committees.

The first option changes nothing about the materials and revisits the adequacy rule. The design's primary prediction is H1b, the active-by-affectedness interaction, with H1a as a supporting prediction. Gating each language on the primary prediction alone, with the passive slope reported as estimation, brings English to 80% on the interaction at 80 participants and Norwegian at 70, at the current verb counts and at the strong threshold. Those crossing points are the smallest samples at which the *lower bound* of the simulated rate reaches 80%, not the first cell whose point estimate happens to. With a few dozen replicates per cell, the two can differ by a whole sample size. Against the 90%

target the recommendation adopts, Norwegian crosses at 80 participants and English at no sample run, its lower bound within the cap peaking at 84%, at 80. No feasible English sample can satisfy the joint gate on both predictions. This option does not help Turkish, whose interaction ceiling is 65% even on its own.

The second option makes the cross-linguistic claim the primary confirmatory test. A hierarchical model fitted to all three languages jointly tests that claim directly, and it multiplies the verb-level information, some 210 verb-by-language units in place of 72, which divides the precision floor by about the square root of three. On the pilot effects, every language's focal terms point the predicted way, so the pooled test starts from a favourable position. Turkish then contributes fully to the confirmatory result, and its language-specific effects are reported as estimates with credible intervals, which reflects what a 72-verb single-language design can support. The pooled design analysis runs on the same machinery used here, and it is now built. Its rationale and the first results are set out below.

The third option expands the verb pool without lengthening any participant's session. Verbs are the scarce resource, and a planned-missingness design assigns each participant a random subset of a larger pool (Graham et al., 2006; Little & Rhemtulla, 2013). Each volunteer still rates about 72 items, so the 45-minute session is unchanged, while the pool grows to, say, 120 to 150 verbs. At about 124 verbs, the English joint ceiling reaches 90%, and the pooled cross-linguistic test gains in proportion. The cost falls on materials, namely affectedness norms and translations for the new verbs. This option is closed. Planned missingness would work if the 72-verb limit constrained the session, but it constrains the pool: the teams are already struggling to find 72 suitable verbs in most languages, so there is no larger set to sample from. That matters beyond this one option, because every ceiling in this report is set by the verb count. With the pool fixed, those ceilings are now permanent features of the design that no materials budget could move, and Turkish, which would need about 348 verbs to reach 80% on its own, is out of reach for good.

The fourth option spends the available volunteer capacity more efficiently. A sequential design analyses the data in waves and stops as soon as the Bayes factor reaches a preset bound or the sample reaches an agreed maximum. This typically needs markedly fewer participants than a fixed-sample design with the same operating characteristics, meaning the same chances of reaching a decisive result (Schönbrodt et al., 2017; Stefan et al., 2019). Optional stopping of this kind poses no problem for Bayes-factor inference (Rouder, 2014).

This option is also closed, and the obstacle is operational. The collecting teams have to put a specific number of participants in front of an ethics committee, and a protocol whose sample size depends on interim results is one that many of them could not explain or defend, however sound it is. Operating characteristics count for nothing if the study cannot be administered by the people running it. The sample size below is therefore a single fixed number, announced in advance.

The recommendation is the first option, with the second kept as the synthesis: each language is tested on its own, gated on the interaction alone, at a Bayes factor of 10, and every team collects 80 participants. The pooled cross-linguistic analysis is preregistered alongside, as the test of the general claim, and its figures are reported below without its being the confirmatory test.

Gating on the interaction alone makes the design work, and the effects themselves were never the problem. When a study has to satisfy both predictions at once, its chances are the two ceilings multiplied together, and the passive-only slope is the one that holds them down. It tops out at 83% in English and 85% in Norwegian however many participants are recruited, while the interaction reaches 97.1% and 99.7%. Dropping the conjunction lifts the English ceiling from 81% to 97.1% and the Norwegian ceiling from 85% to 99.7%. There was little theoretical reason for the conjunction either. The claim CLAPS tests is that affectedness matters *more* for passives than for actives, which is the interaction. The passive-only slope checks that the interaction means what it should, and that belongs in the report as an estimate. It is reported in full below, with its own detection rate, as a prespecified positive control.

The threshold needs justifying in plainer terms than its name suggests, because the Bayes factor used here is directional and does not mean what the conventional verbal labels mean. A directional Bayes factor of k is reached exactly when the posterior probability of the predicted sign reaches $k/(k+1)$, so a Bayes factor of 3 is a posterior sign probability of 0.75, which is 0.67 standard errors from zero and corresponds to a one-sided tail probability of 0.25. A criterion that lenient will not carry a confirmatory claim through review, whatever it is called, and an earlier draft of this report recommended it. A Bayes factor of 6 is 0.14 on the same scale and 10 is 0.091, and matching a one-sided 5% test on the posterior scale would need about 19. The sampling equivalents given under the method are stricter, since the prior's shrinkage has to be overcome first.

The project's starting point was a Bayes factor of 10, which the internal record of decisions carries as the primary criterion for the focal interaction, with 3 as a secondary one, and 10 is the criterion this report recommends. An interim version of this analysis recommended 6, because the pooled cross-language test was then the confirmatory test and 6 was the strongest criterion it came close to meeting. At the feasible cap, the pooled test detects the interaction in 57% [0.47, 0.67] of simulated studies at a Bayes factor of 10, against 84% [0.76, 0.90] at 6, and no amount of recruiting closes that gap, for the reason the rest of this report sets out: the affectedness effect is bounded by the number of verbs. Judged language by language, the design does not need the concession. A Bayes factor of 6 is a one-sided tail of 0.14, a figure reviewers of a registered report will read as lenient, and the single-language tests give up little by holding to 10. At 80 participants, the move from 6 to 10 costs English 96% to 90% and Norwegian 98% to 97%. The pooled test is the one that pays, which is why its role is the synthesis.

The pooled model keeps its place as the preregistered cross-linguistic synthesis, the one analysis that tests the general claim directly and the one whose evidence grows with every language recruited.

Its operating characteristics at three languages are what the simulations give: 57% [0.47, 0.67] at a Bayes factor of 10 and 84% [0.76, 0.90] at 6, combining the pooled cells at or below the cap, since cell by cell the arm carries only 19 to 30 replicates and no single cell settles anything on its own. Those figures are not a power claim for the study as it will be run. The number of languages is not known, and a claim conditional on it would need a simulation with synthetic languages, and no such simulation has been done. They are reported as the sensitivity the synthesis would have with three languages.

Each language is therefore tested on its own against a Bayes factor of 10. English reaches 90% [0.83, 0.95] on the interaction at 80 participants and Norwegian 97% [0.92, 0.99]. The Norwegian figure clears a 90% target by the rule this report uses for adequacy, the exact one-sided lower bound, at 93%. The English figure does not. Its point estimate is at the target, its lower bound is 84%, and over the 480 replicates at 80 participants and above the rate is 89% [0.86, 0.91], so English should be described as powered at about 90%, a figure the simulation cannot certify above the target. Stricter criteria are recorded for reference. At a Bayes factor of 19, a posterior sign probability of 0.95 and the equivalent of a one-sided 5% test on the posterior scale, English reaches 85% and Norwegian 91%. Turkish reaches 57% [0.47, 0.66], and its ceiling on the interaction is 65% at a Bayes factor of 10 however many participants are recruited. Turkish contributes fully to the pooled synthesis and is reported as estimation on its own. That is what 72 verbs buy in a language whose interaction is the weakest of the three. Turkish is also the one language whose passive-only slope is the better-detected of its two focal predictions, 75% against 57% at a Bayes factor of 10. Anyone reporting Turkish on its own should lead on the slope.

Eighty participants is the recommendation, and 100 would not change it. Between those two samples, the simulations differ by less than their own resolution, and the analytic bounds say the most that 20 further participants could add is a few points, because the binding information is in the verbs. Seventy would change it. At 70, the English rate falls to 76%, a drop larger than the Monte-Carlo error, so 80 is the smallest sample at which English sits at the target and the figure has no slack below it. The case for 80 is that it is the number the teams can collect.

What Is Settled and What Is Still Open

The design analysis varies several things at once, and it is worth separating the questions the results have answered from those that remain a matter of judgement. Three findings are settled and no further simulation will overturn them. Power for the affectedness effect is governed by the verb count, so recruitment alone cannot rescue a language that is short of target. English and Turkish do not reach the conventional target on both focal predictions at any feasible per-language sample, whereas Norwegian does. Pooling the three languages multiplies the verb-level information the estimate rests on, and it is the only lever that stays inside the agreed cap without changing the materials, but with three languages its evidence is limited by the variation between them, which neither verbs nor participants reduce.

The choices that follow from this are set out in Table 3. Each row is a decision for the collaborators, with what the evidence says about it and what, if anything, is still outstanding.

Table 3

Design decisions, with what settled each.

Choice	Decision	What settled it
Adequacy rule	The interaction alone	No feasible sample reaches the conjunction of both focal predictions; dropping it lifts the English ceiling from 81% to 97%. The passive-only slope is reported as a positive control
Evidence threshold	Bayes factor of 10	The project's criterion, kept. Per language it costs little: English 96% at 6 against 90% at 10. An interim recommendation of 6 was made for the pooled test, which sits at 57% at 10 and no sample can lift, and that test is no longer the confirmatory one
Power target	90% per language	What the design supports at 72 verbs. Nature Human Behaviour, the target venue, sets its 0.95 requirement for frequentist analysis plans. Of a Bayesian plan it asks either sampling until a Bayes factor of 10, or a stated maximum feasible sample at which an inconclusive result would still matter, and it attaches no power threshold to the second. Norwegian clears 90% by the lower-bound rule, at 93%. English meets it as a point estimate, with a plateau of 89% above 80 participants
Sample size	80 per language	Between 80 and 100 the simulations differ by less than their own resolution, and the verb count binds. At 70 English falls to 76%, so 80 has no slack below it. 80 is the number the teams can collect

Choice	Decision	What settled it
Confirmatory test	Per language, interaction alone	Each language is powered and judged on its own. The pooled model is kept as the preregistered synthesis: with three languages it reaches 57% at a Bayes factor of 10, and its evidence grows with the number of languages, which is not yet known
Turkish	Estimation only	Its ceiling on the interaction is 65% at a Bayes factor of 10 whatever the sample. Settled; further cells will not change it
Verb pool	72 verbs, fixed	Closed by the project: the limit is the supply of suitable verbs, not session length, so the pool cannot be enlarged
Stopping rule	Fixed sample, announced in advance	Closed by the project: teams must take a stated sample size to their ethics committees. The statistical objection to sequential testing does not apply, but the operational one does

None of these now waits on further computation. The project closed two of them, and the simulations settled the rest, having said everything they can about them. The table records which is which. The internal record of decisions still states the success criterion for both focal predictions together, and will be amended to the interaction alone once the collaborators confirm the criterion.

One caveat applies to the whole table. Every rate in this report is computed over all fitted cells, whether or not they pass the convergence screen, since a real analysis would not discard a study for imperfect sampling. The proportion passing that screen falls as the sample grows, so rates at different sample sizes are averages over somewhat different mixtures of converged and unconverged fits. On the screened subset, the ordering of adjacent sample sizes sometimes reverses. This does not affect the recommendation, which rests on the plateau across sample sizes and on a sample size chosen for feasibility, but it is a reason to treat small differences between adjacent cells as noise. At 80 participants, the screen is passed by 27 of 120 English fits, 33 of 120 Norwegian and 12 of 120 Turkish. The failures are on effective sample size and R-hat under the light replicate sampler, with no divergent transitions or treedepth saturations in any of them, and the confirmatory fits use a sampler six times heavier. Over the passing fits alone, the interaction rate is 81% for English and 100% for Norwegian, and the English difference is within what a subset of that size would show by chance.

The Pooled Analysis: Rationale and Expectations

The pooled cross-linguistic design analysis, retained above as the synthesis, has been implemented and validated. Each simulated study draws every language's effects from that language's own pilot posterior, exactly as in the per-language assurance runs, and simulates the three datasets. It then analyses them together with the cross-language model used in the earlier synthesis work, which carries correlated random slopes by participant, by verb and by language, with one amendment to the by-verb term set out below. The evidence criteria are unchanged, namely directional Savage-Dickey Bayes factors on the pooled focal coefficients under the same zero-centred priors. The recommendation is 80 participants per language, with 100 the most any team would be asked for, so the estimate combines the pooled cells at or below 100. The grid itself runs from 50 to 150 per language, and the cells above 100 are retained for record interest. No further simulation is running, and none is scheduled to run. The only CLAPS tasks still in the queue are administratively held remnants of an earlier array that will not execute. The per-language decision arm, at 70, 80 and 100 participants, carries 120 replicates per cell, which brings its Monte-Carlo error near a power of 80% down to about four percentage points. The pooled cells hold far fewer, as the next paragraph says. The pooled analytic ceiling given below is computed from the fitted per-language models, not from a single pooled fit. The pooled model has separately been fitted to the combined real pilot data, and what that fit shows is set out in its own subsection below, because it qualifies the figures reported here.

The pooled arm has finished computing. Of the 180 cells submitted, 135 returned, the rest lost to the ten-day wall, and nothing is still running. Judged on the interaction alone, the pooled test succeeds in 57% [0.47, 0.67] of simulated studies at a Bayes factor of 10, combining the 107 replicates at or below the cap, and in 84% [0.76, 0.90] at a Bayes factor of 6. The first figure is why the confirmatory claim rests on the per-language tests. No single pooled cell settles anything on its own: each holds between 8 and 30 replicates, and their Monte-Carlo errors run to about nine percentage points, so they are combined here. Only 13 of those replicates pass every convergence check, too few to condition on.

The rationale is theoretical before it is statistical. The hypothesis CLAPS tests is cross-linguistic, and the evidence for it has always been read across languages, most explicitly in the meta-analytic synthesis of the five earlier studies (Ambridge et al., 2023). The per-language analyses treat each language as a self-contained confirmatory study, so each inherits its own verb-count ceiling. The pooled model instead tests the general claim directly. Its population-level affectedness slope and interaction are the cross-linguistic effects, and its by-language random slopes let each language depart from them, so generality is itself under test.

Statistically, pooling eases the binding constraint identified above. The precision floor on the affectedness effect is set by the verb-level information, and the pooled analysis draws on about 210 verb-by-language units against 72, which shrinks the floor by roughly the square root of three. The pilot slopes point the predicted way in all three languages, at 0.26, 0.18 and 0.23 logits per one within-language standard deviation of affectedness for the main effect and −0.44, −0.15 and −0.27 on the same scale for the interaction, so their average, which leans on the more precisely estimated languages, sits comfortably clear of zero. Partial pooling also shares strength in the other direction, since each language's estimate borrows from the others in proportion to its own uncertainty, and Turkish gains most from that. The fitted coefficients on the model's Gelman-scaled predictor are about twice these per-standard-deviation values because that predictor has a standard deviation of approximately 0.5.

The expected differences from the per-language results follow from this. First, the ceilings should effectively disappear as constraints. With the pooled effects near the averages above, and a precision floor of about 0.05 logits per standard deviation, both pooled focal ceilings sit near 100%. The pooled test should therefore be limited by ordinary sampling noise alone. That expectation was formed before the pooled cells returned, and the results reported above are what it should be read against: at a Bayes factor of 6 the pooled interaction reaches 84% [0.76, 0.90], at 10 57% [0.47, 0.67], and both are flat in sample size, as a verb-limited design predicts. Second, the limiting quantity passes from the verb count to the variation between languages, especially in the interaction, whose pilot values span −0.44 to −0.15 per standard deviation. That variation does not shrink with more verbs or more participants, and with only three languages its variance is only weakly pinned down by the data and rests partly on the prior (Gelman, 2006), so it is the number to watch in the results and in sensitivity checks. Under the pilot values, it still leaves the pooled ceilings near 97% or higher. That figure is a reference point. It holds the between-language variance fixed at the spread of the three pilot values, as though it were known. It is not known, and with three languages it is barely identified, so the achievable rate is materially lower. Comparing the simulated pooled result against this number therefore overstates any shortfall, and the pooled arm's observed rate should be read against a benchmark that carries the uncertainty in that variance. Third, the interpretation changes. A strong pooled Bayes factor alongside a wide Turkish-specific interval is a coherent and likely outcome, and it is the same pattern the cross-linguistic literature shows. The per-language conclusions above stand unchanged. Norwegian is confirmable on its own, and the English and Turkish per-language joint tests remain capped whatever the pooled result.

The pooled cells have since returned, and they do sit below these expectations at the strongest threshold. The likely cause lies in the between-language variance component, which the two fits to the real pilot data reported next put near 0.6, about twice the spread of the three pilot estimates. The pooled test's sensitivity to the prior on that component has not been examined. It bears on the synthesis alone, leaves the sample size untouched, and is the one check this report leaves open.

The pooled model on real pilot data

The pooled analysis model has been fitted to the combined real pilot data of all three languages, 75,528 ratings over 210 verbs, once in its original form and once under the amendment set out in the next subsection, as a check that it behaves on real ratings before thousands of simulated fits assume it does. The result is reported here in full, because it bears on the figures above in both directions.

The effects come out as expected. The focal interaction is estimated at −0.48, with a 95% credible interval from −1.04 to +0.31 and a posterior probability of 0.91 that it carries the predicted sign, which is a directional Bayes factor of about 10. The unweighted average of the three per-language estimates that the simulation draws from, −0.57, falls comfortably inside that interval, so the data-generating process the design analysis uses is consistent with what the pooled model recovers from real ratings.

The sampling was another matter. The fit did not meet the convergence criteria, saturating the maximum treedepth on 5,968 of its 8,000 post-warm-up iterations, alongside one divergent transition, an R-hat of 1.008 and a minimum bulk effective sample size of 453. Saturation on that scale is a substantial failure. It means the sampler could not traverse the posterior within its step budget, which is the geometry problem the by-verb amendment below was meant to remove, appearing here on real data and more severely than in simulation.

More consequential is the fit's implication for the variation between languages, the quantity that limits the pooled test. Its standard deviation for the focal interaction is estimated at 0.59, with an interval from 0.07 to 1.65, where the design analysis leans on the spread of the three per-language point estimates, 0.30, about half as large. Were the larger value nearer the truth, the pooled test would be less powerful than the figures above report, since the precision of the cross-language average scales with it. Three things pull the other way. The interval is very wide and easily contains the smaller value. The simulation's own procedure induces more between-language spread than the raw 0.30, because each replicate draws each language's effects from its own posterior. And the estimate comes from the poorly sampled fit just described, so it flags the concern without settling the value.

A refit under the amended specification, started on 20 August 2026, returned on 27 August after 6 days and 21 hours, close to the time the original fit took. The effects are unchanged to two decimal places. The focal interaction is −0.48, with a 95% credible interval from −1.01 to +0.30 and a posterior probability of 0.92 that it carries the predicted sign, a directional Bayes factor of about 11, and the standard deviation of the interaction across languages is 0.57, with an interval from 0.09 to 1.59. The larger between-language spread therefore survives the amendment, and with it the qualification above. The sampling improved in one respect only. The divergent transition is gone, but the sampler saturated the maximum treedepth on 6,000 of its 8,000 post-warm-up iterations, with an R-hat of 1.010 and a minimum bulk effective sample size of 393, so the geometry the amendment was meant to remove does not lie in the by-verb term. The likelier seat is the by-language term, whose standard deviations are estimated from three groups and barely identified, the funnel a hierarchical model shows when a group-level variance is left unconstrained by the data. None of this disturbs the sample size, which follows from the verb count and the number of languages, but it bears on the synthesis, as the next subsection sets out.

The by-verb term, and why it is amended

The amended pooled model drops the by-verb random slope on affectedness, keeping `(1 + S_Type | Verb)`, which is the term the single-language models already use. The reason is that affectedness barely varies within a verb. Each verb carries two values, one for each agent gender, and their spread is about 0.6% of the spread between verbs, an SD of 0.18 against 30.9. A random slope estimated from that much variation is close to a second intercept, and the sampler shows it.

The evidence is in the runs already completed, and it is one-sided. Across the 135 completed pooled fits, which carry the by-verb slope, 59% produced at least one divergent transition and 47% saturated the treedepth. Across 360 completed single-language fits at 80 participants, which use the term recommended here, both figures are zero. (These come from the per-cell diagnostic records. The summary tables this report reads do not carry them, so the text quotes them without deriving them.) Divergences and treedepth saturation are signatures of awkward posterior geometry, so they do not go away with longer chains, and divergent transitions can bias the estimates they contribute to. The amendment removes that geometry at the cost of a variance component the data cannot inform.

Two qualifications apply. The operating characteristics reported above were computed under the unamended model, so they describe a specification marginally different from the one preregistered. The dropped component is estimated from almost no information and the focal interaction's precision is dominated by the by-language term, so the figures are expected to carry over with little change, but that expectation has not been simulated and should not be presented as though it had. Separately, the amendment addresses geometry, leaving efficiency untouched, and the two failure modes call for different remedies. Most non-convergence in both model families is a failure to reach the effective-sample-size and R-hat thresholds, which affects the single-language models as much as the pooled one and follows from fitting 8,000 draws two thousand times over. The confirmatory analysis is fitted once and takes the heavier sampler the proposal sets for confirmatory fits, the sixteen chains of 5,000 iterations given above, six times the post-warm-up draws of a replicate and enough to clear those thresholds. Treedepth saturation is the other mode, and a longer run does nothing for it, because the binding constraint is the depth of each trajectory. The amendment exists to remove it, and the refit was the test of whether it succeeds. It does not. On the real pilot data, the amended model saturated the treedepth on three quarters of its iterations, as the unamended one had. The proposal leaves no discretion on the consequence. Zero divergences and zero treedepth saturations are locked acceptance criteria for confirmatory fits, and a fit that misses them falls to the next level of the model ladder, its result set aside. Raising the maximum tree depth is not an available remedy, the configuration reserving it for diagnosis. For the pooled model, the next level is the uncorrelated cross-language model, which the record of decisions names for the pooled design analysis and which no pooled cell has simulated. Its operating characteristics are expected to be close to those reported above, since the focal interaction's precision is set by the by-language term and hardly by the correlations, but that is an expectation and has not been simulated. This is a further reason the pooled analysis is preregistered as the synthesis, with the confirmatory claim carried by the per-language tests, whose fits at 80 participants show no divergences and no saturations in any of the 360 replicates.

Limitations and Status

The status differs across the two analyses. The literature-anchored cross-check is complete for all three languages and points to a modest sample at the full verb set. The data-grounded analysis, which is the primary basis, is substantially complete. Norwegian is finished and is adequately powered at about 100 participants. English and Turkish fall short of the joint power target at every sample size run. The finding that participants alone cannot bring English and Turkish to target rests on the affectedness effect being verb-limited. The 400-participant cells confirm that for English at 32 replicates, and the analytic ceilings bound it from above. No further simulation bears on the sample size: recruitment is capped at 100 participants, and every sample at or below that cap has been simulated to full replication. No fit is outstanding either, the refit described above having returned on 27 August.

That status statement concerns the joint criterion, which this report no longer recommends. Gated on the interaction alone at a Bayes factor of 10, Norwegian clears a 90% target within the cap and English sits at it, with a plateau just below. Turkish remains short, and permanently so.

Two aspects of the simulation record qualify how far these figures can be pressed. Each assurance replicate reuses one of the first 24 to 40 saved posterior draws, a block at the head of the chain. Checked against the full posterior, those blocks are representative for English and Norwegian and sit slightly below the mean for the Turkish affectedness slope, so the simulated Turkish figures are, if anything, mildly pessimistic. The English blocks also run lower than the decision arm at the two sample sizes both cover: 31 of 48 older replicates detect the interaction at 70 and 100 participants against 202 of 240 in the decision arm, a gap chance would not readily produce, so the mixed figures this report quotes at 70 and 100 are, if anything, slightly conservative for English. The 80-participant cell is decision arm only and unaffected. (These counts, like the diagnostic ones quoted earlier, are read from the per-cell records.) The analytic ceilings use all 8,000 draws and are unaffected, and the pooled cross-linguistic design analysis and the decision arm sample their draw indices randomly over the full pilot posteriors, so the limitation does not reach them. Replication itself is complete, since the original single-language grid finished all 720 of its cells and the low counts noticed mid-run were cells still computing.

An audit of seed allocation, prompted by a failing continuous-integration check, found two grids sharing a seed base and a smoke grid sharing a seed with a production cell. Both are fixed. Neither touched a reported figure, because the two arms concerned produced no output at all, their arrays having been cancelled before any task started. The arm the recommendation rests on was checked directly. Its 1,080 cells carry 1,080 distinct seeds, contiguous over 940000 to 941079, disjoint from every other committed grid once the per-language offsets are expanded, and each resolves to its own output file.

The analysis rests on 4,328 simulated studies in the cells reported here, and they are not evenly spread. The per-language assurance grid holds 1,814 and the literature-anchored cross-check 2,019, against 135 in the pooled arm, which is 3% of the total. The synthesis rests on the smallest of them, and that has a consequence. Combining the pooled cells at or below the cap gives 57% [0.47, 0.67] for the interaction at a Bayes factor of 10 and 84% [0.76, 0.90] at 6. The single-language figures rest on 120 replicates per cell at the decision sample sizes and the pooled one does not: it is the best estimate available of the synthesis's sensitivity at three languages, and its power in the study as it will be run remains undemonstrated.

That estimate carries its own uncertainty, and the scale of it can be stated. Taking the pooled cells at face value under a flat prior, the true rate at a Bayes factor of 10 is most likely near 57%, with an exact interval from 47% to 67% and a probability of less than 0.01 that it is as high as 70%. A run of 380 further cells at 80 participants per language was submitted to tighten this and cancelled before any of them started, and then set aside on the strength of its own arithmetic. The chance of landing more than ten points from the current figure is about 0.06, and the new figure would have fallen within five points of the current one roughly 65% of the time, so the run would have measured more precisely a rate that no plausible value brings to the target the confirmatory tests are held to. Several weeks of computing to firm up a number that changes no decision is not a good trade.

The more substantial reason is that the pooled arm does not appear to have much room to improve. Its rate should be read against a benchmark that treats the between-language variance as itself estimated from only three languages, and on that footing the observed rate is close to what three languages permit. The constraint is the number of languages, which the project cannot change, and it would survive any respecification of the model. Further simulation would therefore measure this design more precisely without producing a better one, and the design analysis is closed on that basis.

Individual cells close to the threshold should be read as near target, since each rests on 22 to 144 simulated studies, and the conclusions rest on the pattern across sample sizes.

Every power figure in this report is directional. The Bayes factor weighs the predicted sign against the opposite one, and no null hypothesis enters it (Morey & Wagenmakers, 2014). Nature Human Behaviour asks for a factor of at least 10 in favour of the experimental hypothesis over the null. It also asks authors to state the distribution representing the theory's prediction, and it names a half-normal on the likely effect size as one acceptable form. Stage 1 will declare that form. At the effects behind the rates above, and with the alternative as a half-normal on the predicted side, the interaction's factor against a point null is about 11 for English, 51 for Norwegian and 2 for Turkish, the ordering the directional criterion also gives. English clears 10 by little. A simulated detection rate under that criterion cannot be recovered from the stored records, which keep only the directional factor, and a design analysis is specific to the criterion it targets (Schönbrodt & Wagenmakers, 2018).

This is an a-priori power analysis, a simulation under assumed effects to size the study, and it is distinct from the confirmatory analysis of the real ratings that will be run once data collection is complete. The analysis code and configuration are openly available, and the report rebuilds from the committed summary tables without the raw data, which remain private to CLAPS until the project reaches a more advanced stage. The simulation programme is now closed, so this report is finalised on the evidence described, with no further cells pending. The one check left open, the pooled arm's sensitivity to the prior on the between-language variance, concerns the synthesis and has no bearing on the sample size.

References

Albers, C., & Lakens, D. (2018). When power analyses based on pilot data are biased: Inaccurate effect size estimators and follow-up bias. *Journal of Experimental Social Psychology*, *74*, 187–195. https://doi.org/10.1016/j.jesp.2017.09.004

Ambridge, B., Arnon, I., & Bekman, D. (2023). He was run-over by a bus: Passive – but not pseudo-passive – sentences are rated as more acceptable when the subject is highly affected. New data from Hebrew, and a meta-analytic synthesis across English, Balinese, Hebrew, Indonesian and Mandarin. *Glossa Psycholinguistics*, *2*(1). https://doi.org/10.5070/G6011177

Ambridge, B., Bidgood, A., Pine, J. M., Rowland, C. F., & Freudenthal, D. (2016). Is passive syntax semantically constrained? Evidence from adult grammaticality judgment and comprehension studies. *Cognitive Science*, *40*(6), 1435–1459. https://doi.org/10.1111/cogs.12277

Aryawibawa, I. N., & Ambridge, B. (2018). Is syntax semantically constrained? Evidence from a grammaticality judgment study of Indonesian. *Cognitive Science*, *42*(8), 3135–3148. https://doi.org/10.1111/cogs.12697

Barr, D. J., Levy, R., Scheepers, C., & Tily, H. J. (2013). Random effects structure for confirmatory hypothesis testing: Keep it maximal. *Journal of Memory and Language*, *68*(3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001

Beavers, J. (2011). On affectedness. *Natural Language & Linguistic Theory*, *29*(2), 335–370. https://doi.org/10.1007/s11049-011-9124-6

Bidgood, A., Pine, J. M., Rowland, C. F., & Ambridge, B. (2020). Syntactic representations are both abstract and semantically constrained: Evidence from children's and adults' comprehension and production/priming of the English passive. *Cognitive Science*, *44*(9), e12892. https://doi.org/10.1111/cogs.12892

Brysbaert, M., & Stevens, M. (2018). Power analysis and effect size in mixed effects models: A tutorial. *Journal of Cognition*, *1*(1), 9. https://doi.org/10.5334/joc.10

Bürkner, P.-C., & Vuorre, M. (2019). Ordinal regression models in psychology: A tutorial. *Advances in Methods and Practices in Psychological Science*, *2*(1), 77–101. https://doi.org/10.1177/2515245918823199

Button, K. S., Ioannidis, J. P. A., Mokrysz, C., Nosek, B. A., Flint, J., Robinson, E. S. J., & Munafò, M. R. (2013). Power failure: Why small sample size undermines the reliability of neuroscience. *Nature Reviews Neuroscience*, *14*(5), 365–376. https://doi.org/10.1038/nrn3475

Clark, H. H. (1973). The language-as-fixed-effect fallacy: A critique of language statistics in psychological research. *Journal of Verbal Learning and Verbal Behavior*, *12*(4), 335–359. https://doi.org/10.1016/S0022-5371(73)80014-3

Darmasetiyawan, I. M. S., & Ambridge, B. (2022). Syntactic representations contain semantic information: Evidence from Balinese passives. *Collabra: Psychology*, *8*(1), 33133. https://doi.org/10.1525/collabra.33133

Dowty, D. R. (1991). Thematic proto-roles and argument selection. *Language*, *67*(3), 547–619. https://doi.org/10.1353/lan.1991.0021

Gelman, A. (2006). Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, *1*(3), 515–534. https://doi.org/10.1214/06-BA117A

Gelman, A., & Carlin, J. (2014). Beyond power calculations: Assessing Type S (sign) and Type M (magnitude) errors. *Perspectives on Psychological Science*, *9*(6), 641–651. https://doi.org/10.1177/1745691614551642

Graham, J. W., Taylor, B. J., Olchowski, A. E., & Cumsille, P. E. (2006). Planned missing data designs in psychological research. *Psychological Methods*, *11*(4), 323–343. https://doi.org/10.1037/1082-989X.11.4.323

Ioannidis, J. P. A. (2008). Why most discovered true associations are inflated. *Epidemiology*, *19*(5), 640–648. https://doi.org/10.1097/EDE.0b013e31818131e7

Judd, C. M., Westfall, J., & Kenny, D. A. (2017). Experiments with more than one random factor: Designs, analytic models, and statistical power. *Annual Review of Psychology*, *68*, 601–625. https://doi.org/10.1146/annurev-psych-122414-033702

Lewandowski, D., Kurowicka, D., & Joe, H. (2009). Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis*, *100*(9), 1989–2001. https://doi.org/10.1016/j.jmva.2009.04.008

Liddell, T. M., & Kruschke, J. K. (2018). Analyzing ordinal data with metric models: What could possibly go wrong? *Journal of Experimental Social Psychology*, *79*, 328–348. https://doi.org/10.1016/j.jesp.2018.08.009

Little, T. D., & Rhemtulla, M. (2013). Planned missing data designs for developmental researchers. *Child Development Perspectives*, *7*(4), 199–204. https://doi.org/10.1111/cdep.12043

Liu, L., & Ambridge, B. (2021). Balancing information-structure and semantic constraints on construction choice: Building a computational model of passive and passive-like constructions in Mandarin Chinese. *Cognitive Linguistics*, *32*(3), 349–388. https://doi.org/10.1515/cog-2019-0100

Morey, R. D., & Wagenmakers, E.-J. (2014). Simple relation between Bayesian order-restricted and point-null hypothesis tests. *Statistics & Probability Letters*, *92*, 121–124. https://doi.org/10.1016/j.spl.2014.05.010

Perugini, M., Gallucci, M., & Costantini, G. (2014). Safeguard power as a protection against imprecise power estimates. *Perspectives on Psychological Science*, *9*(3), 319–332. https://doi.org/10.1177/1745691614528519

Pinker, S., Lebeaux, D. S., & Frost, L. A. (1987). Productivity and constraints in the acquisition of the passive. *Cognition*, *26*(3), 195–267. https://doi.org/10.1016/S0010-0277(87)80001-X

Rouder, J. N. (2014). Optional stopping: No problem for Bayesians. *Psychonomic Bulletin & Review*, *21*(2), 301–308. https://doi.org/10.3758/s13423-014-0595-4

Schönbrodt, F. D., & Wagenmakers, E.-J. (2018). Bayes factor design analysis: Planning for compelling evidence. *Psychonomic Bulletin & Review*, *25*(1), 128–142. https://doi.org/10.3758/s13423-017-1230-y

Schönbrodt, F. D., Wagenmakers, E.-J., Zehetleitner, M., & Perugini, M. (2017). Sequential hypothesis testing with Bayes factors: Efficiently testing mean differences. *Psychological Methods*, *22*(2), 322–339. https://doi.org/10.1037/met0000061

Stefan, A. M., Gronau, Q. F., Schönbrodt, F. D., & Wagenmakers, E.-J. (2019). A tutorial on Bayes factor design analysis using an informed prior. *Behavior Research Methods*, *51*(3), 1042–1058. https://doi.org/10.3758/s13428-018-01189-8

Taylor, J. E., Rousselet, G. A., Scheepers, C., & Sereno, S. C. (2023). Rating norms should be calculated from cumulative link mixed effects models. *Behavior Research Methods*, *55*(5), 2175–2196. https://doi.org/10.3758/s13428-022-01814-7

Vasishth, S., Mertzen, D., Jäger, L. A., & Gelman, A. (2018). The statistical significance filter leads to overoptimistic expectations of replicability. *Journal of Memory and Language*, *103*, 151–175. https://doi.org/10.1016/j.jml.2018.07.004

Veríssimo, J. (2021). Analysis of rating scales: A pervasive problem in bilingualism research and a solution with Bayesian ordinal models. *Bilingualism: Language and Cognition*, *24*(5), 842–848. https://doi.org/10.1017/S1366728921000316

Wagenmakers, E.-J., Lodewyckx, T., Kuriyal, H., & Grasman, R. (2010). Bayesian hypothesis testing for psychologists: A tutorial on the Savage-Dickey method. *Cognitive Psychology*, *60*(3), 158–189. https://doi.org/10.1016/j.cogpsych.2009.12.001

Westfall, J., Kenny, D. A., & Judd, C. M. (2014). Statistical power and optimal design in experiments in which samples of participants respond to samples of stimuli. *Journal of Experimental Psychology: General*, *143*(5), 2020–2045. https://doi.org/10.1037/xge0000014

Appendix A

Composition of the Pilot Sample

This appendix describes the pilot sample from which the affectedness ratings and the descriptive trends in Figure 1 are drawn, and against which the data-grounded analysis fits its models. The counts are read directly from the harmonised pilot data.

Table 4

Composition of the Harmonised Pilot Sample, by Language

Language	Participants	Verbs	Sentence types	Items	Judgements
English	70	72	3	144	30,240
Turkish	70	72	3	144	30,240
Norwegian	57	66	2	132	15,048

Note. Counts are read from the harmonised pilot data. Items are verb-by-agent-gender combinations, so the item count is twice the verb count. Norwegian lacks a pseudo-passive, hence only two sentence types.

Appendix B

Key Code for the Simulation and the Models

This appendix reproduces the central code, lightly abridged, from the analysis scripts. The code shown is the data-generating process for the literature-anchored cross-check, in `R/06_simulate_design.R`, together with the analysis model in `R/04_model_formulas.R` and the Bayes factor in `R/05_hypothesis_tests.R`. The data-grounded analysis uses the same analysis model and Bayes factor, but draws its data-generating effects, random-effect covariance and thresholds from the fitted pilot models, and reads each verb's affectedness from the pilot data. Its assurance and safeguard variants are in `R/10_simulate_from_pilot.R` and `R/10_simulate_from_pilot_v2.R`.

Each simulated dataset is drawn from the same cumulative-logit model that is later fitted. In the cross-check, affectedness is drawn once per verb across a plausible range, which makes it a verb-level predictor, and the ordinal response follows from the cumulative-logit linear predictor.

```
# DGP for the literature-anchored cross-check, abridged from
 ↪  R/06_simulate_design.R.
verb_semantics <- runif(n_verbs, min = -0.5, max = 0.5)   # one value per verb

# For participant pid, verb vid and sentence type st:
sem   <- verb_semantics[[vid]]
b_int <- if (st == "Active")               beta_active_interaction * sem
         else if (st == "Pseudo_Passive") beta_pseudo_interaction * sem
         else 0
eta   <- part_intercepts[pid] + part_slopes[pid] * sem +
         verb_intercepts[vid] + beta_semantics * sem + b_int

cum_probs <- plogis(thresholds - eta)        # seven ordered categories
probs     <- diff(c(0, cum_probs, 1))
resp      <- sample(seq_len(7), size = 1, prob = probs)
```

The analysis model is the maximal random-effects structure consistent with the design, fitted with the cumulative-logit family.

```
# Analysis model, from R/04_model_formulas.R.
brms::bf(
  Response ~ S_Type * Semantics_scaled +
```

```
    (1 + S_Type * Semantics_scaled | Participant) +
    (1 + S_Type | Verb),
  family = brms::cumulative(link = "logit", threshold = "flexible")
)
```

Evidence for each prediction is a directional Savage-Dickey Bayes factor, with the prior sampled alongside the posterior and the posterior mass taken in the predicted direction.

```
# Directional Savage-Dickey Bayes factor, from R/05_hypothesis_tests.R.
# post_vals and prior_vals are draws of the focal coefficient; the negative
# direction is shown, as for the primary prediction H1b.
post_prob  <- mean(post_vals  < 0)
prior_prob <- mean(prior_vals < 0)
bf_10      <- (post_prob / (1 - post_prob)) / (prior_prob / (1 - prior_prob))
```

Appendix C

Full Bayes-Factor Operating Characteristics

This appendix reports the operating characteristics for all four predictions in every cell of the corrected literature-anchored run holding at least ten replicates, set out in Table 5, which covers English, Norwegian and Turkish. The note under the table defines each column. Under Hyp, H1b is the active-by-affectedness interaction, H1a the passive affectedness slope, and H2a and H2b the secondary pseudo-passive interaction in each direction.

Table 5

Operating Characteristics by Design Condition and Prediction

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	40	50	H1a	74%	96%	28	100%	50
English	40	50	H1b	76%	88%	21	100%	50
English	40	50	H2a	46%	74%	7.7	100%	50
English	40	50	H2b	0%	0%	0.13	100%	50
English	40	60	H1a	94%	100%	53	100%	50
English	40	60	H1b	80%	96%	37	100%	50
English	40	60	H2a	68%	94%	25	100%	50
English	40	60	H2b	0%	0%	0.041	100%	50
English	40	70	H1a	76%	98%	31	98%	50
English	40	70	H1b	86%	96%	80	98%	50
English	40	70	H2a	60%	86%	16	98%	50
English	40	70	H2b	0%	2%	0.063	98%	50
English	40	80	H1a	80%	98%	26	98%	50
English	40	80	H1b	96%	98%	180	98%	50
English	40	80	H2a	52%	88%	11	98%	50
English	40	80	H2b	0%	0%	0.093	98%	50
English	40	90	H1a	88%	100%	49	98%	50
English	40	90	H1b	96%	100%	510	98%	50
English	40	90	H2a	58%	88%	12	98%	50
English	40	90	H2b	0%	0%	0.081	98%	50
English	40	100	H1a	78%	94%	97	96%	50
English	40	100	H1b	94%	100%	760	96%	50
English	40	100	H2a	42%	82%	8.6	96%	50
English	40	100	H2b	0%	0%	0.12	96%	50
English	40	120	H1a	84%	96%	62	88%	50
English	40	120	H1b	98%	98%	3,200	88%	50
English	40	120	H2a	68%	82%	20	88%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
English	40	120	H2b	0%	0%	0.049	88%	50
English	72	50	H1a	98%	100%	760	92%	50
English	72	50	H1b	98%	100%	860	92%	50
English	72	50	H2a	58%	84%	15	92%	50
English	72	50	H2b	0%	0%	0.067	92%	50
English	72	60	H1a	100%	100%	1,600	90%	50
English	72	60	H1b	98%	100%	1,200	90%	50
English	72	60	H2a	72%	92%	19	90%	50
English	72	60	H2b	0%	2%	0.052	90%	50
English	72	70	H1a	98%	100%	580	84%	50
English	72	70	H1b	98%	100%	5,900	84%	50
English	72	70	H2a	60%	86%	15	84%	50
English	72	70	H2b	0%	0%	0.066	84%	50
English	72	80	H1a	96%	100%	710	82%	50
English	72	80	H1b	100%	100%	7,800	82%	50
English	72	80	H2a	78%	94%	55	82%	50
English	72	80	H2b	0%	2%	0.018	82%	50
English	72	90	H1a	100%	100%	1,800	82%	50
English	72	90	H1b	100%	100%	980,000	82%	50
English	72	90	H2a	80%	88%	58	82%	50
English	72	90	H2b	0%	0%	0.017	82%	50
English	72	100	H1a	96%	100%	940	92%	50
English	72	100	H1b	100%	100%	970,000	92%	50
English	72	100	H2a	78%	90%	51	92%	50
English	72	100	H2b	0%	0%	0.019	92%	50
English	72	120	H1a	100%	100%	480	90%	50
English	72	120	H1b	100%	100%	990,000	90%	50
English	72	120	H2a	86%	94%	56	90%	50
English	72	120	H2b	0%	0%	0.018	90%	50
Norwegian	40	50	H1a	78%	94%	26	100%	50
Norwegian	40	50	H1b	62%	86%	19	100%	50
Norwegian	40	60	H1a	68%	90%	38	96%	50
Norwegian	40	60	H1b	90%	96%	64	96%	50
Norwegian	40	70	H1a	72%	94%	27	100%	50
Norwegian	40	70	H1b	88%	98%	110	100%	50
Norwegian	40	80	H1a	76%	90%	40	96%	50
Norwegian	40	80	H1b	96%	100%	210	96%	50
Norwegian	40	90	H1a	82%	94%	34	98%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
Norwegian	40	90	H1b	96%	100%	280	98%	50
Norwegian	40	100	H1a	86%	96%	49	98%	50
Norwegian	40	100	H1b	94%	100%	850	98%	50
Norwegian	40	120	H1a	86%	100%	35	98%	50
Norwegian	40	120	H1b	100%	100%	1,400	98%	50
Norwegian	72	50	H1a	92%	98%	550	98%	50
Norwegian	72	50	H1b	94%	98%	340	98%	50
Norwegian	72	60	H1a	96%	100%	440	96%	50
Norwegian	72	60	H1b	100%	100%	570	96%	50
Norwegian	72	70	H1a	100%	100%	700	96%	50
Norwegian	72	70	H1b	100%	100%	2,000	96%	50
Norwegian	72	80	H1a	100%	100%	410	94%	50
Norwegian	72	80	H1b	100%	100%	8,000	94%	50
Norwegian	72	90	H1a	98%	100%	1,100	84%	50
Norwegian	72	90	H1b	100%	100%	490,000	84%	50
Norwegian	72	100	H1a	96%	100%	580	88%	50
Norwegian	72	100	H1b	100%	100%	990,000	88%	50
Norwegian	72	120	H1a	98%	100%	1,100	90%	50
Norwegian	72	120	H1b	100%	100%	990,000	90%	50
Turkish	40	50	H1a	74%	95%	62	100%	19
Turkish	40	50	H1b	79%	95%	130	100%	19
Turkish	40	50	H2a	26%	74%	6.4	100%	19
Turkish	40	50	H2b	0%	0%	0.16	100%	19
Turkish	40	60	H1a	74%	90%	73	100%	50
Turkish	40	60	H1b	86%	96%	79	100%	50
Turkish	40	60	H2a	54%	74%	11	100%	50
Turkish	40	60	H2b	2%	2%	0.089	100%	50
Turkish	40	70	H1a	88%	98%	54	98%	50
Turkish	40	70	H1b	90%	98%	59	98%	50
Turkish	40	70	H2a	58%	86%	13	98%	50
Turkish	40	70	H2b	0%	0%	0.075	98%	50
Turkish	40	80	H1a	80%	96%	43	98%	50
Turkish	40	80	H1b	94%	98%	180	98%	50
Turkish	40	80	H2a	70%	94%	31	98%	50
Turkish	40	80	H2b	0%	0%	0.032	98%	50
Turkish	40	90	H1a	90%	94%	67	86%	50
Turkish	40	90	H1b	98%	98%	400	86%	50
Turkish	40	90	H2a	62%	82%	21	86%	50

(continued)

Language	Verbs	N	Prediction	P(BF>=10)	P(BF>=3)	Median BF	Convergence	Reps
Turkish	40	90	H2b	0%	0%	0.052	86%	50
Turkish	40	100	H1a	88%	96%	44	100%	50
Turkish	40	100	H1b	98%	100%	980	100%	50
Turkish	40	100	H2a	70%	86%	30	100%	50
Turkish	40	100	H2b	0%	2%	0.033	100%	50
Turkish	72	50	H1a	96%	98%	1,800	90%	50
Turkish	72	50	H1b	96%	100%	880	90%	50
Turkish	72	50	H2a	58%	86%	16	90%	50
Turkish	72	50	H2b	0%	0%	0.064	90%	50
Turkish	72	60	H1a	100%	100%	480	94%	50
Turkish	72	60	H1b	96%	100%	1,000	94%	50
Turkish	72	60	H2a	62%	92%	21	94%	50
Turkish	72	60	H2b	0%	0%	0.049	94%	50
Turkish	72	70	H1a	96%	100%	410	88%	50
Turkish	72	70	H1b	100%	100%	8,000	88%	50
Turkish	72	70	H2a	70%	90%	26	88%	50
Turkish	72	70	H2b	0%	0%	0.038	88%	50
Turkish	72	80	H1a	100%	100%	1,300	98%	50
Turkish	72	80	H1b	100%	100%	970,000	98%	50
Turkish	72	80	H2a	68%	86%	24	98%	50
Turkish	72	80	H2b	0%	0%	0.041	98%	50
Turkish	72	90	H1a	96%	100%	750	94%	50
Turkish	72	90	H1b	100%	100%	970,000	94%	50
Turkish	72	90	H2a	68%	88%	33	94%	50
Turkish	72	90	H2b	0%	0%	0.031	94%	50
Turkish	72	100	H1a	96%	96%	1,400	82%	50
Turkish	72	100	H1b	100%	100%	990,000	82%	50
Turkish	72	100	H2a	88%	96%	49	82%	50
Turkish	72	100	H2b	0%	0%	0.02	82%	50
Turkish	72	120	H1a	100%	100%	550	88%	50
Turkish	72	120	H1b	100%	100%	990,000	88%	50
Turkish	72	120	H2a	84%	100%	78	88%	50
Turkish	72	120	H2b	0%	0%	0.013	88%	50

Note. The probability columns give the proportion of simulated studies reaching a Bayes factor of 10 and of 3, Median BF their median to two significant figures, Convergence the proportion of fits meeting the criteria, and Reps the number of studies in the cell. Cells with fewer than ten replicates are excluded, and Norwegian has no pseudo-passive, hence half as many rows.

Appendix D

Guide to the Analysis Code

This appendix maps the analysis code, which is openly available at https://github.com/pablobernabeu/CLAPS-analysis. It is meant to make the repository navigable for a reader who wants to check how a number in this report was produced, or to reuse part of the machinery. The raw pilot data are not published, since they are participant-level records held privately until the project is further advanced, so the repository ships the simulation and analysis code together with the aggregated summaries the report reads.

Three entry points cover most purposes. A reader checking a reported number wants `outputs/`, which holds the aggregated summaries and is where every figure in this report comes from. A reader checking how a simulated study is built and analysed wants `R/06_simulate_design.R` for the literature-anchored generator, `R/10_simulate_from_pilot_v2.R` for the pilot-grounded one, and `R/05_hypothesis_tests.R` for the Bayes factor. A reader intending to rerun the analysis wants `scripts/`, whose files are numbered in the order they are meant to run, together with the matching submitters in `hpc/`. Table 6 sets out what each top-level directory holds.

Table 6

Layout of the public analysis repository.

Location	What it holds
`R/`	The building blocks: data validation and preprocessing (`01`, `02`), priors and model formulas (`03`, `04`), the Bayes factor (`05`), the two simulation engines (`06`, and `10` with `11`), convergence diagnostics (`07`) and summarising (`08`)
`scripts/`	Runnable steps, numbered in running order, from harmonising the pilot (`00`) through fitting the pilot models, generating a design grid, running one cell, to aggregating results (`aggregate_pilot.R`, `aggregate_pooled.R`, `aggregate_literature.R`)
`hpc/`	One SLURM submitter per arm. Each takes a grid file and an output directory, and runs one grid row per array task
`config/`	The design grids, one row per simulated condition, plus the model and prior configuration
`outputs/`	The aggregated summaries this report reads, and the figure it displays
`tests/`	Unit tests for the validation, prior, diagnostic and Bayes-factor logic
`reports/`	This report, its bibliography and citation style

The unit of work is the cell, meaning one simulated condition: a language, a participant count, a verb count and a replicate index. A grid file in `config/` lists the cells, an array task runs one row, and each cell is saved as a single `.rds` file holding its summary, its Bayes factors and its convergence diagnostics. The aggregation scripts read a directory of those files and reduce them to the tables in `outputs/`. Reading one saved cell is therefore the quickest way to see exactly what the pipeline computes per simulated study.

Two conventions are worth knowing before reading the code. Names carrying `pilot` refer to the data-grounded analysis that conditions on the fitted pilot models and forms the primary basis of this report, while names carrying `literature` refer to the cross-check that conditions on the published effect sizes. Separately, the suffix `_v2` on a simulation engine marks an extension of the original, never a replacement for it, so both files are live and the pair should be read together.

Three files in `R/` generate simulated studies, `06`, `10` and `11`, and between them they implement only two engines. The engines differ in where the true effects come from, and the second of them is written at two scopes.

The literature-anchored engine is `R/06_simulate_design.R`. It invents a study from assumed effect sizes taken from the published estimates, drawing participants, verbs and affectedness values from scratch. Because its parameters are stipulated, extending it from one language to several is only a matter of adding a language level to the same recipe, so this engine covers both scopes in one file: `simulate_claps_data()` for a single language and `simulate_claps_data_multilanguage()` beside it.

The pilot-grounded engine is `R/10_simulate_from_pilot.R`. It takes the true effects from a model already fitted to the pilot data, so a simulated study inherits that language's estimated slopes, thresholds and variance components together with its real verbs and their measured affectedness. `R/10_simulate_from_pilot_v2.R` loads that file and adds the assurance and safeguard variants, which sample the effects afresh from the pilot posterior at each replicate, where the base version holds them at its mean.

Extending this second engine across languages is not a matter of adding a level, because each language carries its own fitted model. A pooled study has to be assembled from three separate sets of parameters, simulated language by language and then combined before a single cross-language model is fitted to the result. That orchestration is what `R/11_simulate_pooled_v2.R` adds. It loads the `_v2` file, calls its per-language simulator once for each language, binds the three datasets and fits the pooled model, so it is the same engine at a wider scope.

The files therefore stack, each loading the one below it and adding to it: `11` on `10_v2` on `10`, and separately `R/06_simulate_design_gelman.R` on `06`. That last file is the narrowest of them. It changes only the scale on which the analysed predictor is expressed, leaving the simulated responses identical, so that the literature-anchored cells are analysed on the same footing as the confirmatory model.

Both engines end in the same place. Each defines a cell runner, `run_design_cell()` in `06` and `run_databased_cell_v2()` and `run_pooled_cell_v2()` in `10_v2` and `11`, which simulates one dataset, fits it with `R/04_model_formulas.R` and the priors in `R/03_define_priors.R`, computes the Bayes factors with `R/05_hypothesis_tests.R` and records the diagnostics from `R/07_extract_diagnostics.R`, then writes the `.rds` for that cell. Only the source of the true effects differs; the model, the evidence criterion and the convergence standard are shared, which makes the two analyses comparable. Table 7 summarises where each engine's effects come from and which file implements it at each scope.

Table 7

The two simulation engines and the files implementing them.

Engine	Where the effects come from	Single language	All three pooled
Literature-anchored	Published effect sizes, stipulated	`06`	`06`, the same file
Pilot-grounded	Estimated from the fitted pilot models	`10`, extended by `10_v2`	`11`, which loads `10_v2`

Following one arm end to end is the quickest way to see how the pieces connect. For the pooled analysis reported here, the grid is `config/design_grid_pooled_v2.csv`, one row per simulated study; `hpc/submit_pooled_v2_array.sh` submits one array task per row; `scripts/07_pooled_cell_v2.R` runs a single row by calling `run_pooled_cell_v2()`; and `scripts/aggregate_pooled.R` reduces the resulting `.rds` files to `pooled_power_pilot.csv` in `outputs/design_summary_pilot/`, which is the table this report reads for every pooled figure it quotes.